# Nearest neighbour interpolation

**1**

**a**

**i** $8\text{ mm}$ **ii** $17\text{ mm}$ **iii** $7\text{ mm}$ **iv** $13.5\text{ mm}$

**b** At $(0, 0)$ since we would expect it to rain $17\text{ mm}$ there, whereas as at $(3.5, -4)$ we would expect it to rain $12.5\text{ mm}$ since it is on the edge between the regions for $C$ and $D$.

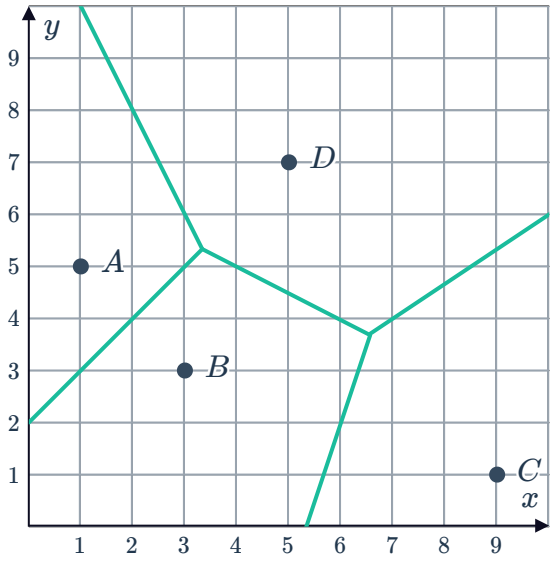
**2**

**a** $18^\circ\text{C}$

**b** $18^\circ\text{C}$

**3**

**a**




**b**

**i** Yes **ii** No **iii** Yes **iv** No

**c**

**i** No **ii** Yes **iii** Yes **iv** Yes

**4** Point chosen should be in region $E$ and within $5$ units of the point $(-3, -4)$.
Example integer answers: $(-1, -1), (-1, 0), (0, 0), (0, -1), (0, -2), (1, -1), (1, -2), (1, -3)$

**5**

**a** $0.25$ inches **b** $0.27$ inches **c** $0.26$ inches **d** $0.43$ inches

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

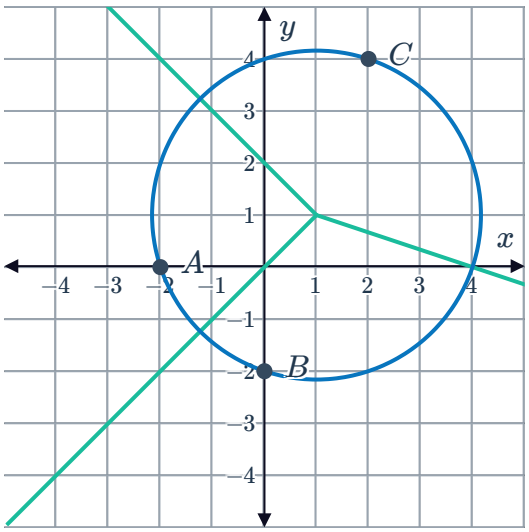
## Largest empty circle

**6**

**a**

**i** Centre: $(1, 1)$, radius: $3.16$

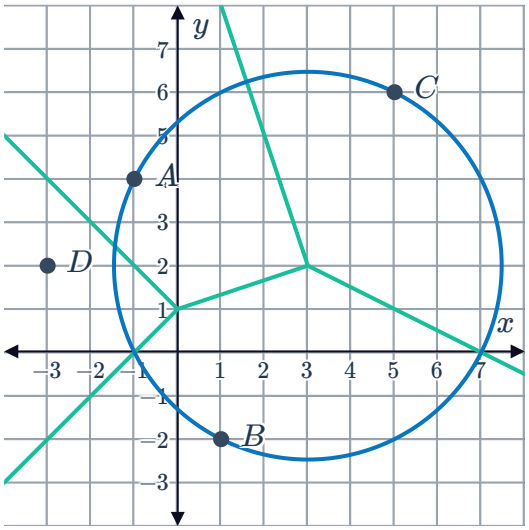
**ii**




**b**

**i** Centre: $(3, 2)$, radius: $4.47$

**ii**



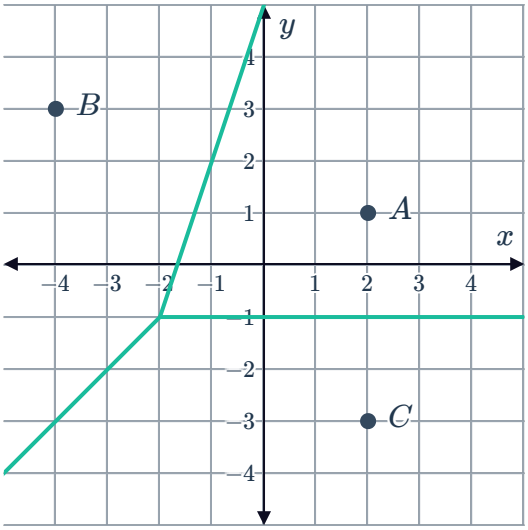

**7** Centre: $(-3, 5)$ or $(5, -3)$, radius: $2\sqrt{10}$

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

**8**

**a**

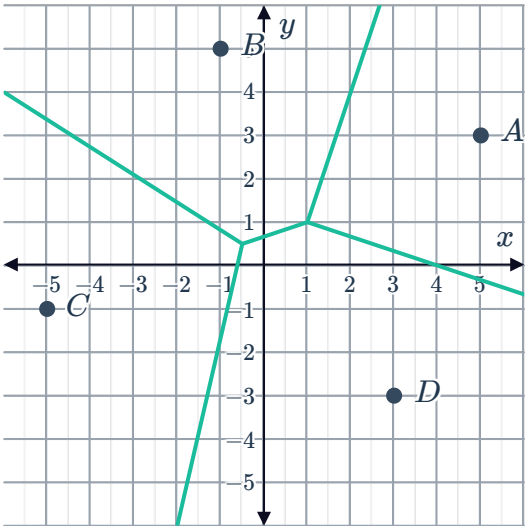
**i**




**ii** Centre: $(-2, -1)$, radius: $4.472$

**b**

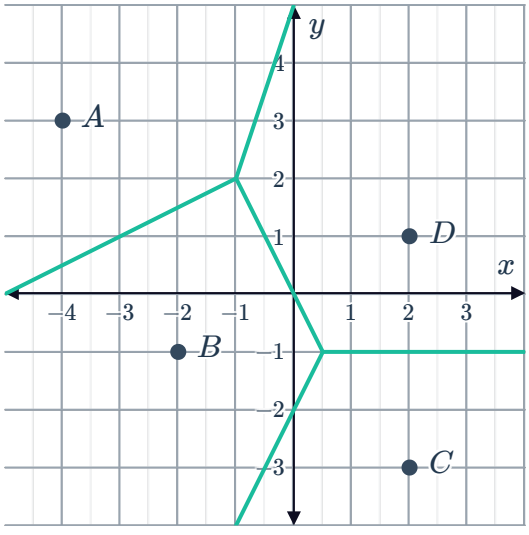
**i**




**ii** Centre: $(-0.5, 0.5)$, radius: $4.528$

**9**

**a**




**b** $(-1, 2)$

© ClipSAT: By Mr. Mohamed Abdallah +966597688647